

BANNUSHA SHAIK

Software Engineer · Python / FastAPI · Backend Systems · GenAI Applications

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SUMMARY

Software Engineer intern (B.Tech AIML, 2027) who ships working systems — APIs, data models, caches, realtime, CI/CD, and monitoring — not just tutorials. Fortune 500 internship at SKF delivering a production enterprise app. Built a search service at 1,000+ QPS with sub-10ms p95, a realtime collaborative editor (Operational Transform, 100+ sessions), and an LLM SQL generator with JWT auth and SSE streaming. Seeking Software Engineer Intern / SDE Intern / Backend Engineer (0–1 year) roles in Bangalore or remote India.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, HTML/CSS

Backend: FastAPI, Node.js, Express, REST APIs, API Design, Asynchronous Programming, WebSockets, Server-Sent Events (SSE), JWT, Authentication, Object-Oriented Programming, Data Structures & Algorithms

Data: PostgreSQL, Redis, MySQL, MongoDB, Schema Design, Query Optimization, Indexing, Caching, Information Retrieval, BM25, Inverted Index

DevOps: Git, GitHub, Docker, Docker Compose, CI/CD, GitHub Actions, Linux, Nginx, Unit Testing (pytest), Postman, Agile / SDLC

Observability: Prometheus, Grafana, Alertmanager, Loki, Metrics, Logging, Dashboards, Alerting

AI / GenAI: Large Language Models (LLMs), Google Gemini API, Prompt Engineering, Generative AI, NLP, scikit-learn, Pandas, NumPy

Enterprise: Microsoft Power Apps, SharePoint, Office 365, Power BI

EXPERIENCE

Software and Digital Developer Intern — SKF (Fortune 500, Industrial Technology)

Jan 2025 – Present

Bangalore, India · On-site · Production enterprise software (Power Platform)

- Shipped a production 5S Audit Management app on Microsoft Power Platform (Power Apps, SharePoint, Office 365) that replaced paper factory audits — digital checklists, photo evidence, scoring, and action tracking used by manufacturing teams.
- Designed an 11-screen canvas app across all five 5S pillars: per-pillar question banks, department/zone selection, identity-aware user context, and weighted qualification scoring with qualifying thresholds.
- Modeled SharePoint-backed data on the SKF O365 tenant (audit drafts, zone lists, write-back via Patch) and built score-card / compliance-trend views so factory leadership can see bottlenecks.
- Worked on-site with shop-floor stakeholders — requirements, usability, and rollout in a live industrial environment. Owned delivery from data model to deployment readiness.

GenAI Data Analytics Job Simulation — Tata Group (Forage)

2024

Labeled as a job simulation, not an internship

- Job simulation (Forage): modeled delinquency risk on structured financial datasets using GenAI-assisted analytics and wrote executive-ready recommendations.

PROJECTS

NEXORA — Production Search Engine

Python · FastAPI · Redis · BM25

github.com/bannushaxddd/NEXORA · nexora-1kjq.onrender.com

- Built a search service at 1,000+ QPS with sub-10ms p95: custom Okapi BM25 over a hand-built inverted index (no search library), Redis cache (~85% hit), async REST APIs for search, corpus, and live analytics.
- Benchmarked ~2ms cached / ~5ms uncached; pytest + pytest-asyncio suite; OpenAPI docs. Demonstrates information retrieval, caching, and API design — the same primitives used in production search and RAG pipelines.

SYNCOUT — Realtime Collaborative Code Editor

Python · FastAPI · WebSockets · Docker

github.com/bannushaxddd/SYNCOUT · syncout-x.onrender.com

- Implemented Operational Transform from scratch for conflict-free concurrent editing: 100+ sessions, sub-50ms sync, live cursors, session isolation, built-in chat. Dockerized with GitHub Actions CI/CD.

Observability Stack

Prometheus · Grafana · Alertmanager · Loki · Docker Compose

github.com/bannushaxddd/prometheus-grafana-stack

- End-to-end monitoring: Prometheus scrape configs, Grafana dashboards, Alertmanager routing, Loki logs, custom app /metrics (counters, histograms, gauges), alerting rules and runbooks. Five instrumented services.

GETYOQUERY — Natural Language to SQL

Node.js · Express · React · Gemini · SSE · JWT

github.com/bannushaxddd/GETYOQUERY

- Full-stack GenAI app: NL → SQL across 8 dialects, Gemini streaming over SSE, JWT + bcrypt auth, PostgreSQL query history, parameterized queries, hallucination-reduction prompt patterns.

EDUCATION

B.Tech, Artificial Intelligence & Machine Learning — PES College of Engineering, Bangalore

Expected May 2027

Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, Object-Oriented Programming, Generative AI, Machine Learning, Deep Learning, NLP, Statistics & Probability, Linear Algebra, Cloud Computing.

CERTIFICATIONS

Generative AI: Prompt Engineering Basics — IBM · GenAI Powered Data Analytics Job Simulation — Tata Group (Forage)